

CampusConnect System Documentation

Comprehensive Architectural Specifications, Database Schema, Visual UML Diagrams & Core Workflows

Executive Summary

CampusConnect is a centralized, web-based Progressive Web App (PWA) tailored exclusively for university ecosystems. The system integrates student marketplace transactions, peer service directories, career/internship opportunities, campus event tracking, direct peer-to-peer messaging, promotional notifications, peer referral mechanisms, a seller/provider rating system, and an administrative announcements hub.

Backend Infrastructure: Firebase is utilized entirely as the serverless backend, leveraging **Firebase Authentication**, **Cloud Firestore**, **Cloud Storage for Firebase**, **Cloud Functions**, and **Firebase Cloud Messaging (FCM)**.

1. Requirements & System Specifications

1.1 Architecture & Interfaces

- **User Interface:** Responsive PWA supporting Desktop, Tablet, and Mobile viewports (320px to 2560px).
- **Backend Infrastructure (Firebase Suite):**
 - **Firebase Authentication:** Restricts registration strictly to university domains (.edu .cm or approved institutional domains).
 - **Cloud Storage for Firebase:** Stores media assets, portfolio files, and PDF resumes.
 - **Cloud Firestore:** Real-time NoSQL database managing users, products, services, internships, events, chats, reviews, referrals, and announcements.
 - **Firebase Cloud Messaging (FCM) & Cloud Functions:** Handles promotional push notifications, real-time message alerts, automated referral tracking, and system update broadcasts.

1.2 Functional Requirements (FRD)

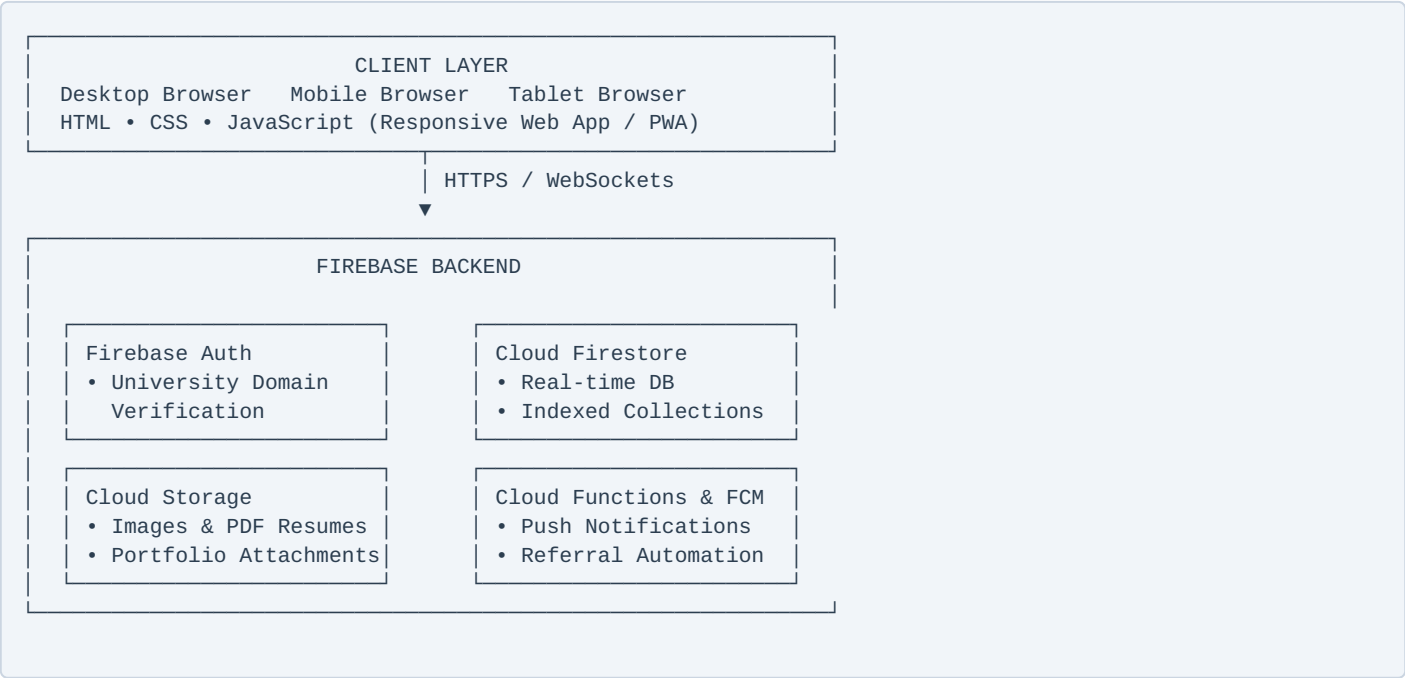
Req ID	Module	Description	Priority
FR-AUTH-01	User Auth	Restrict registration strictly to valid university email addresses (.edu .cm or institutional domain).	High
FR-AUTH-02	User Auth	Send automated verification email upon registration.	High
FR-MKT-01	Marketplace	Create, edit, soft-delete, and mark items as "Sold" in Firestore.	High
FR-MKT-02	Marketplace	Attach up to 5 high-resolution images per listing.	Medium
FR-SRV-01	Services	Post skill listings with hourly rates, skill tags, and portfolio links.	High
FR-JOB-01	Opportunities	Enable verified recruiters/admins to post internships and jobs with deadlines.	High
FR-JOB-02	Opportunities	Allow students to upload PDF resumes and apply directly.	High
FR-EVT-01	Campus Events	Schedule events with location, date, time, category, and capacity flags.	Medium
FR-MSG-01	Messaging	Enable direct, real-time messaging between prospective buyers/clients and sellers/providers.	High

Req ID	Module	Description	Priority
FR-SCH-01	Search/Filter	Search globally or filter by category, price, condition, and posting date.	High
FR-NOTIF-01	Notifications	Send promotional push notifications to inform users about new listings and important updates.	High
FR-REF-01	Growth & Referral	Generate unique invite links/codes allowing students to invite peers and expand platform adoption.	Medium
FR-REV-01	Rating & Feedback	Allow users to submit star ratings and written feedback for sellers and service providers.	High
FR-ANC-01	Announcements	Broadcast administrative announcements and key campus updates on a dedicated board.	Medium

1.3 Non-Functional Requirements (NFRD)

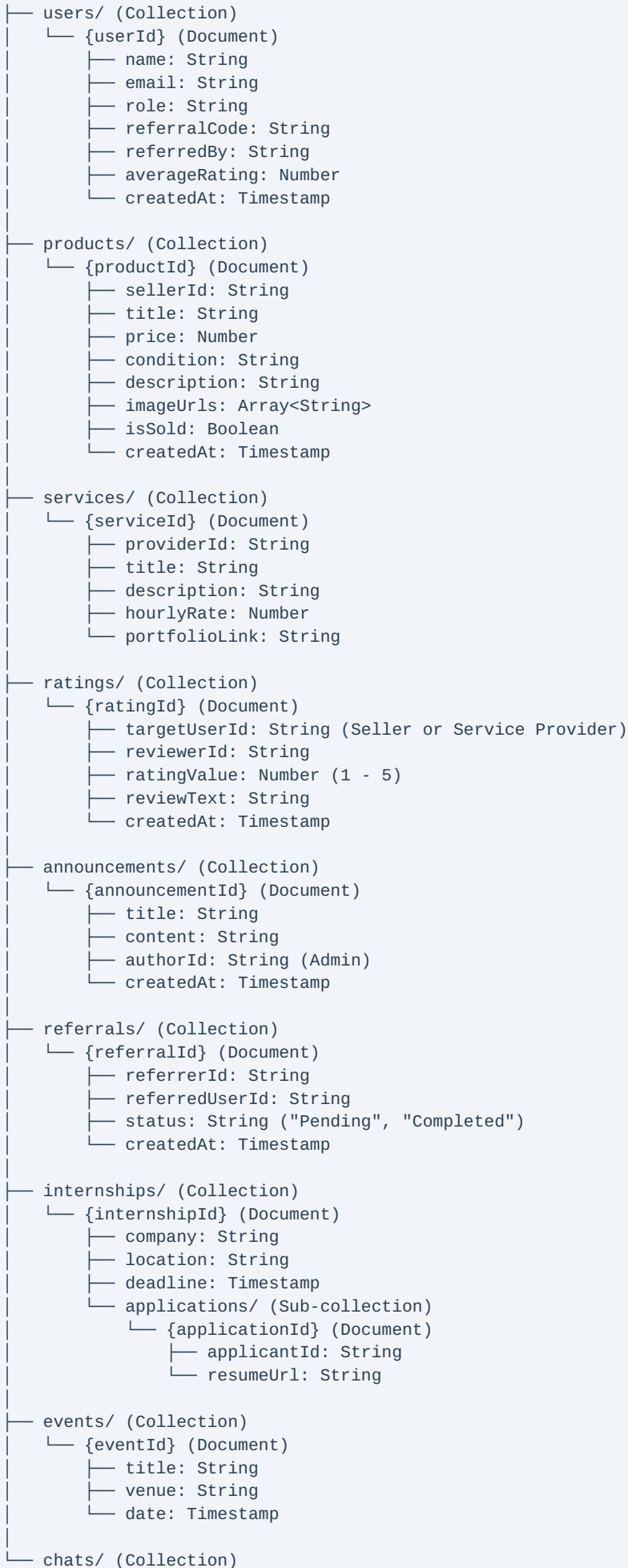
- **Performance:** Initial page load under 2.0 seconds over 4G; search execution under 500ms using indexed Firestore queries.
- **Security:** TLS 1.3/HTTPS transport encryption; granular Firebase Security Rules for documents and collections; media MIME validation.
- **Reliability & Availability:** Multi-region automated replication via Firebase maintaining 99.5%+ uptime.
- **Usability:** Compliance with WCAG 2.1 Level AA accessibility standards.

2. Updated Firebase System Architecture & Software Components



3. Cloud Firestore NoSQL Data Schema

Firestore Root



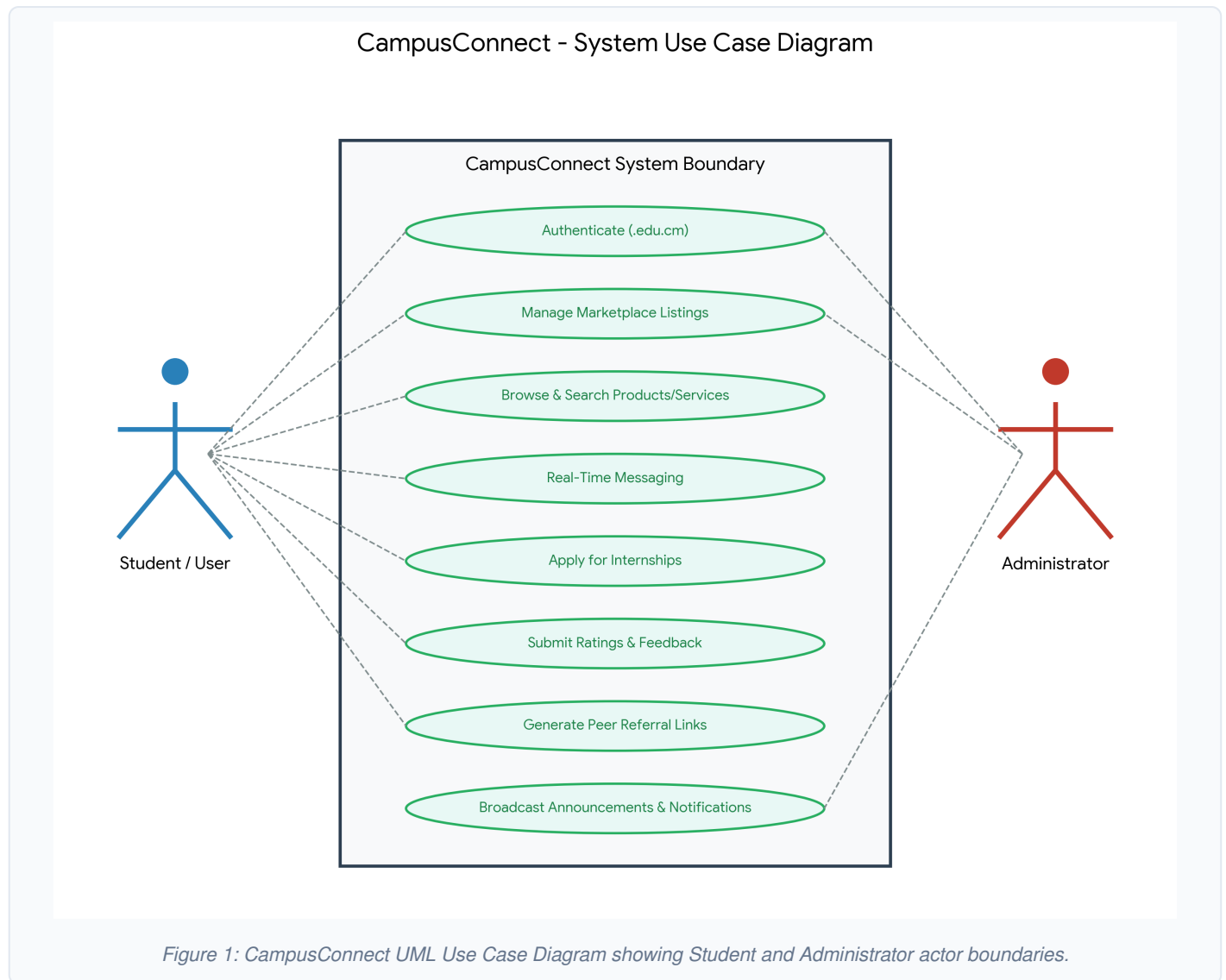
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└─ {chatId} (Document)
  └─ participants: Array<String>
  └─ messages/ (Sub-collection)
    └─ {messageId} (Document)
      └─ senderId: String
      └─ content: String
      └─ timestamp: Timestamp

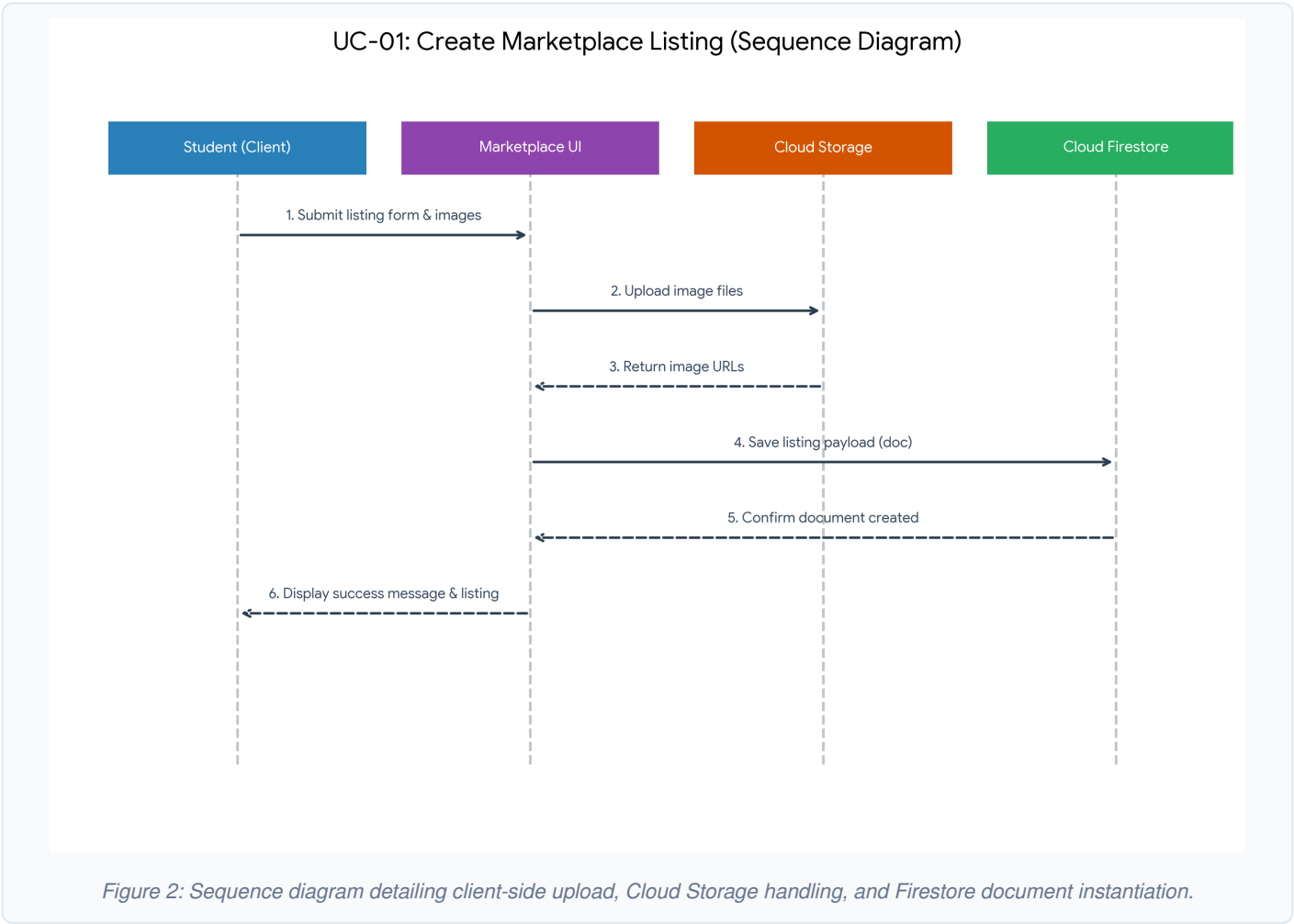
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4. Visual Diagrams: UML Models & System Workflows

4.1 Visual UML Use Case Diagram

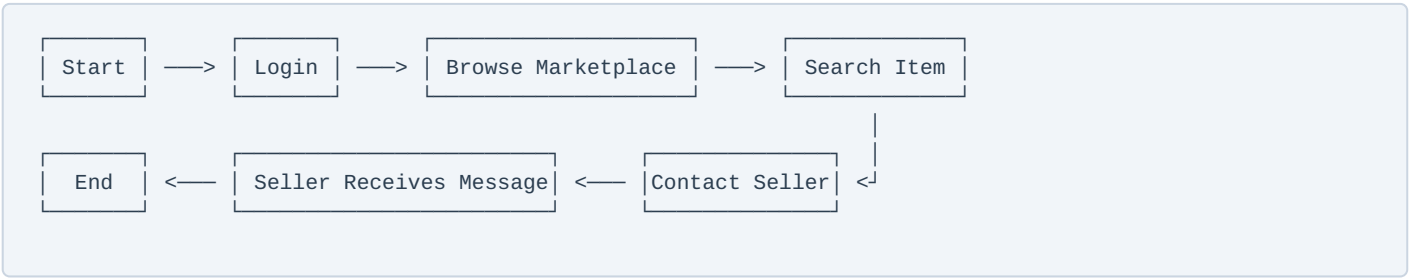


4.2 Visual Sequence Diagram (UC-01: Create Marketplace Listing)

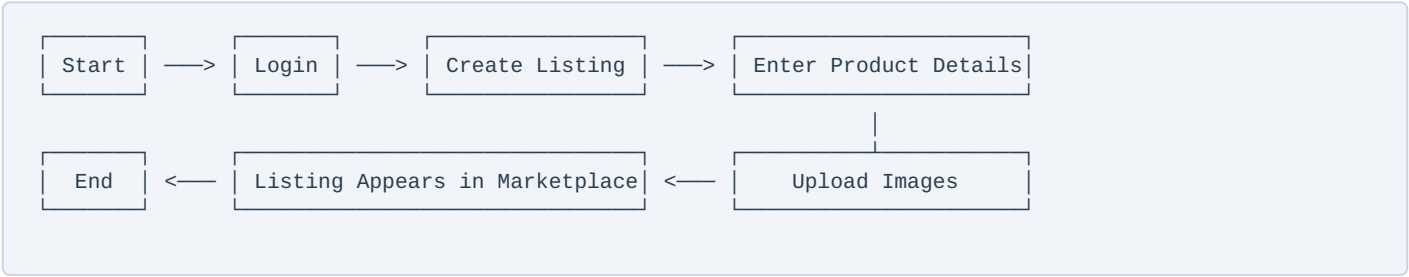


4.3 Core Workflows

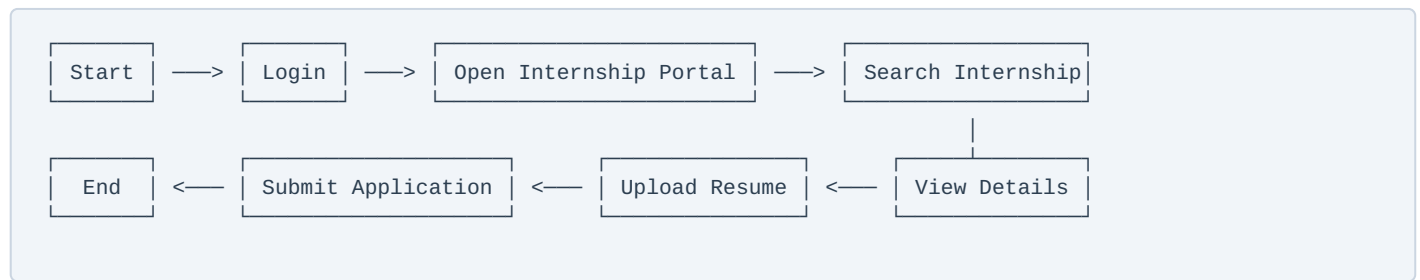
Workflow 1: Buying an Item Flow



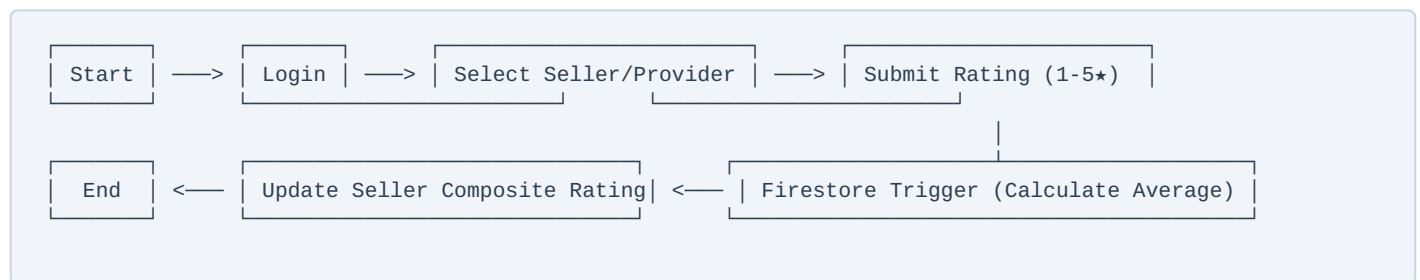
Workflow 2: Selling an Item Flow



Workflow 3: Finding & Applying for an Internship Flow



Workflow 4: Rating & Feedback Workflow



5. Requirement Traceability Matrix (RTM)

REQ Ref ID	Business Objective / User Story	Functional REQ ID	Use Case Ref	Verification Method
REQ-01	Filter marketplace listings by category & price	FR-MKT-01	UC-01	Integration Testing, UI Automated Tests
REQ-02	Direct messaging between buyer and seller	FR-MSG-01	UC-02	Firestore Real-Time Listener Test
REQ-03	Dynamic search across listings with low latency	FR-SCH-01	UC-02	Firestore Composite Index Test
REQ-04	Create peer service page and portfolio links	FR-SRV-01	UC-01 (Variant)	Functional System Test
REQ-05	Apply for internships using PDF resumes	FR-JOB-01, FR-JOB-02	UC-01 (Variant)	Cloud Storage & Firestore UAT
REQ-06	System Security & Restricted Email Auth	FR-AUTH-01, FR-AUTH-02	All Use Cases	Firebase Security Rules & Auth Test
REQ-07	Receive promotional push notifications for new listings & news	FR-NOTIF-01	FCM Notification Flow	FCM Delivery & Mobile Alert Test
REQ-08	Invite other students via referral links to grow platform	FR-REF-01	Referral Generation Flow	End-to-End Invite & Registration Test
REQ-09	Rate and review sellers/providers to build trust	FR-REV-01	Rating & Review Flow	Firestore Trigger & Rating Calculation Test
REQ-10	Broadcast system updates via dedicated announcements section	FR-ANC-01	Announcement View Flow	Admin Content Publishing Test